

Name:

Admission No.

Indian Institute of Technology (Indian School of Mines) Dhanbad

Department of Mining Engineering

Quiz - Winter Semester 2023-2024 for 4th Year B.Tech. + 1st Year Ph.D. in Mining Engineering (36 nos.)

Subject: Deep Coal Mining (Code: MND 412)

Time: 60 minutes (12:00 PM – 01:00 PM)

Date: 18-04-2024

Maximum Marks: 40

*Answer **all** the questions and **marks** are assigned against each question*

1. Describe the procedure of estimation of powered roof support rating. (3 marks)

Step 1: Identification of Weak Parting Planes:

Step 2: Formation of Rock Beds

Step 3: Identification of Caving Layers

Modified empirical-cum-statistical relation for estimating the convergence slope and Effective support resistance:

2. Define paste as a suitable material which is used in backfilling. (2 marks)

Paste is defined as a mixture of material, which akin to toothpaste, comprising 75–85% solid by weight and 1 wt% to 12 wt% binder, containing at least 15 wt% solid fraction $<20\ \mu\text{m}$, when poured into voids the backfill does not bleed water, does not settle in the pipeline, and has a slump $<250\ \text{mm}$.

3. Discuss the different challenges faced by Adriyala Longwall Project of Singareni Collieries Company Limited, Telangana. (2 marks)

Gate roads Drivage: Cable bolts installed of 6.1 m of 60 tons capacity, workability of Bolter Miner (BM) due to its steep gradient and stone cutting in the floor. Use newly supplied Road Headers (RH) for drivage of roadways to avoid surprises with BM as RH technology

Strata Control Management during development:

Cavities occurred in gateroad ways driven in bottom section taking middle clay in to working section. Later geological mapping and lithological study was done and working section has been changed by leaving middle clay in the overlying strata with 1 m coal underneath it, which improved the stability of gate roadways.

The cut throughs were driven in between two gate roadways at 200 m interval. Initially cut throughs were driven across the major horizontal stress direction, in which severe strata control problems were encountered and cavities also taken place. Later the direction of cut throughs has been changed and aligned with major horizontal stress direction which eliminated major strata problems.

Initially gate roadways were supported with roof bolting with linked wire mesh in the roof. Later rigid wire mesh has been introduced in place of linked wire mesh to reduce supporting cycle time and to improve the supporting efficiency of the system. The roof conditions and drivage rates have been improved with introduction of rigid wire mesh.

Laying of Civil foundations on loose overburden for conveyor belts

Transport of Shearer:

Provision of Air Chilling Plant:

Dealing with Goaf Water:

Spontaneous heating in LWP-2 salvage:

4. A longwall panel 6,200 ft long with a face length of 820 ft is planned for a seam 1,300 ft deep and 16.5 ft thick. Mining is full seam height. Assume UK conditions apply. Find:

(22 marks)

- (a) The maximum subsidence that will develop during the life of the panel

Solution:

$$\frac{S}{h} = 0.9 \left\{ 1 - e^{\left[-2.47 \left(\frac{x}{H} \right)^2 \right]} \right\} = 0.9 \left\{ 1 - e^{\left[-2.47 \left(\frac{820}{1300} \right)^2 \right]} \right\}$$

$$S = 0.5631 * 16.5 = 9.29 \text{ ft}$$

- (b) The maximum subsidence after the face has progressed 520 ft.

Solution:

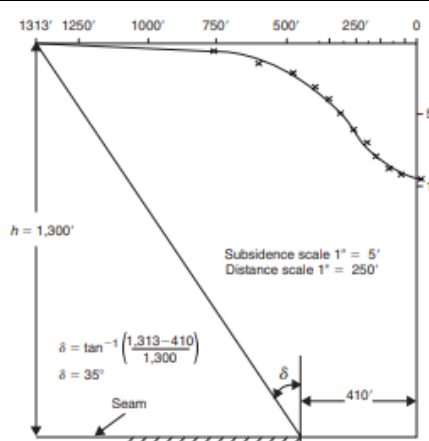
$$\frac{S'}{S_{max}} = \left\{ 1 - e^{\left[-2.70 \left(\frac{x}{H} \right)^2 \right]} \right\} = \left\{ 1 - e^{\left[-2.70 \left(\frac{520}{1300} \right)^2 \right]} \right\}$$

$$S = 9.29 * 0.3507 = 3.25 \text{ ft}$$

- (c) The surface subsidence profile when fully developed.

Solution:

s/S_{max}	0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95	1.00
l/h	1.01	0.59	0.47	0.37	0.31	0.27	0.23	0.19	0.16	0.13	0.09	0.06	0
s	0	0.50	0.99	1.98	2.97	3.96	4.95	5.94	6.93	7.92	8.91	9.41	9.90
l	1313	767	611	481	403	351	299	247	208	169	117	78	0



e., vertical to horizontal scale ratio is 5:250

Table 8.3 Peak strain multipliers^a

W/H	E+	E-
0.10	0.25	2.40
0.20	0.52	2.25
0.30	0.70	2.00
0.40	0.77	1.73
0.50	0.80	1.35
0.60	0.75	1.06
0.70	0.68	0.86
0.80	0.65	0.70
0.90	0.65	0.60
1.00	0.65	0.55
1.10	0.65	0.54
1.20	0.65	0.53
1.30	0.65	0.52
1.40	0.65	0.51

Note

a Tension (+), compression (-).

- (d) The surface strain profile; include specific values for E_t , E_c , e_c .

Solution:

$$\frac{W}{H} = \frac{820}{1300} = 0.631$$

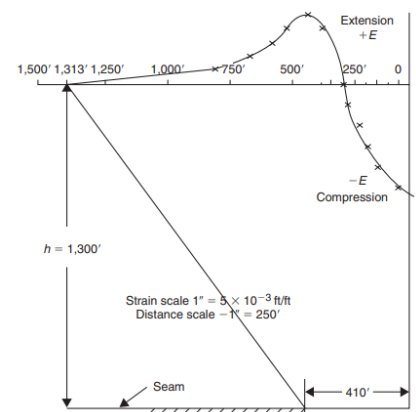
$$\text{For } +E = \frac{x - 0.8}{0.65 - 0.8} = \frac{0.631 - 0.5}{0.8 - 0.5}$$

$$+E = 0.73$$

$$\text{For } -E = \frac{x - 1.35}{0.7 - 1.35} = \frac{0.631 - 0.5}{0.8 - 0.5}$$

$$-E = 1.07$$

$$E_t = 0.73 * \frac{S}{H} = 0.73 * \frac{9.29}{1300} = 5.2166 \times 10^{-3}$$



$$E_c = 1.07 * \frac{S}{H} = 1.07 * \frac{9.29}{1300} = 7.6463 \times 10^{-3}$$

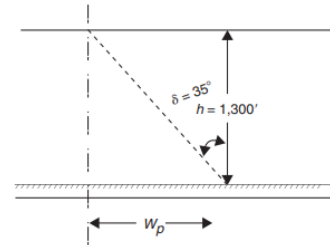
$$\frac{-e}{-E} = e^{\left[-4\left(\frac{W}{H}-0.5\right)^2\right]} = e^{\left[-4\left(\frac{820}{1300}-0.5\right)^2\right]} = 0.93388$$

$$e_c = E_c * 0.93388 = 7.6463 \times 10^{-3} * 0.93388 = 7.14 \times 10^{-3}$$

	Extension (+E)							Compression (-E)					
e/E	0.0	0.2	0.4	0.6	0.8	1.0	0.8	0	0.2	0.4	0.6	0.8	1.00
l/h	1.01	0.63	0.52	0.45	0.41	0.34	0.30	0.23	0.21	0.18	0.15	0.12	0.05
$e \times 10^{-3}$	0	1.13	2.25	3.38	4.51	5.64	4.51	0	1.66	3.32	4.98	6.64	8.30
l	1313	819	676	585	533	442	390	299	273	234	195	156	65

- (e) The width of a barrier pillar needed to protect a ventilation shaft that serves the mine.

Solution: $W_p = H * \tan(\delta) = 1300 * \tan(35) = 910.26 \text{ ft}$



- (f) The time required to mine the panel at 7,600 tons per shift, 2 shifts per day, 250 days per year.

Solution: Total panel volume = $(6,200) * (820) * (16.5) = 83,88,600 \text{ ft}^3$

6200 ft= 1889.76 m

820 ft= 249.93 m

16.5 ft= 5.03 m

@ 7,600 Tons/Shift; 2 shifts/day; 250 days/year;

Assuming $20 \text{ ft}^3/\text{ton}$ for coal i.e. $\gamma = 100 \text{ lbf/ft}^3$

Volume moved in a year = $(7,600) * (2) * (250) * (20) = 7,60,00,000 \text{ ft}^3/\text{year}$ to move $83.886 \times 10^6 \text{ ft}^3$ it will take 1.1 years.

- (g) The average rate of advance per day

Solution: Cubic feet moved per day = $(7,600) * (2) * (20) = 304 \times 10^3 \text{ ft}^3$

Face advance/day = $\frac{\text{cubic ft/day mined}}{\text{Face Area}} = \frac{304 * 10^3}{820 * 16.5} = 22.5 \text{ ft}$

Average face advance/day = 22.5 ft

- (h) The maximum subsidence after mining four identical panels adjacent to the first panel and trough width.

Solution: Now the face width = $(5) * (820) = 4,100 \text{ ft}$

Using Figure 3 SEH – P-9

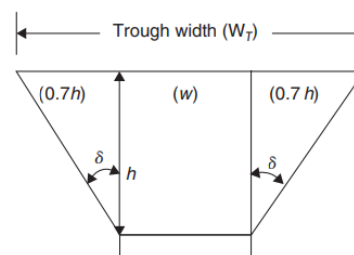
Face width = $\frac{4100}{3.281} = 1250 \text{ m}$

Depth = 396 m

$\Rightarrow \frac{S_{max}}{m} = 0.9$

$\Rightarrow S_{max} = (0.9) * (16.5) = 14.9 \text{ ft}$

$\Rightarrow \delta = 35^\circ$



$$\Rightarrow W_T = 2 * (1300) * (\tan 37) + 4,100 = 6059 \text{ ft}$$

- (i) The maximum subsidence and trough width after mining an identical panel 80 ft directly below the first panel.

Solution: Panel 1 $S_{max} = 9.9 \text{ ft}$, from (a).

Subsidence due to Panel 2

Depth = 1397 ft = 426 m

W = 820 ft = 250 m

$$\frac{S}{h} = 0.9 \left\{ 1 - e^{\left[-2.47 \left(\frac{x}{H} \right)^2 \right]} \right\} = 0.9 \left\{ 1 - e^{\left[-2.47 \left(\frac{820}{1397} \right)^2 \right]} \right\} = 0.57301$$

$$S = 16.5 * 0.57301 = 9.4547 \text{ ft}$$

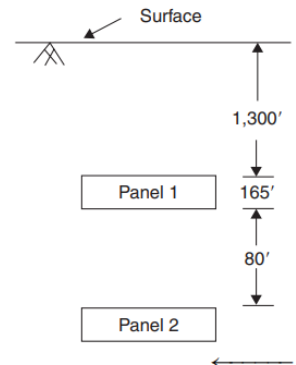
$$\Rightarrow \frac{S_{max}}{m} = 0.55$$

$$S_{max} = (0.55) * (165) = 9.1 \text{ ft}$$

$$S_{max}(\text{Total}) = 9.9 + 9.1 = 19 \text{ ft} = 5.8 \text{ m}$$

- (j) Assuming US conditions with a subsidence factor of 0.65 and an angle of draw of 28° , find the maximum subsidence and trough width for a single panel and for four adjacent panels

Solution: Assuming Utah conditions



$$\frac{S_{max}}{m} = 0.65; \delta = 28^\circ @ \text{critical width}$$

Single panel

$$W_c = 2H \tan(28) = 2 * 1300 * 0.5317 = 1382 \text{ ft}$$

A single panel is only 820 ft i.e. sub-critical

If sub-critical width.

$$S_{max}(\text{Utah}) = \frac{0.65}{0.9} \left(\frac{S_{max}(\text{UK})}{m} \right) m = \frac{0.65 * 9.29}{0.9} = 6.71 \text{ ft}$$

$$\text{Trough width } (W_T) = 2 \tan(28) (1300) + 820 = 2202 \text{ ft}$$

Four Adjacent panels.

Panel width = (5) * (820) = 4,100 $\Rightarrow$ supercritical

$$S_{max} = 0.65 * 16.5 = 10.73 \text{ ft}$$

$$W_T = 2[\tan(28)] * 1300 + 5 * 820 = 5482 \text{ ft}$$

- (k) Estimate the maximum tensile and compressive strains for a single panel assuming US conditions (part j).

Solution: For Utah conditions, single panel. $\frac{S_{max}}{H} = \frac{7.16}{1300} = 5.508 \times 10^{-3} \text{ ft/ft}$

$$\Rightarrow +E = (0.73)(5.508 \times 10^{-3}) = 4.021 \times 10^{-3} \text{ ft/ft}$$

$$-E = (1.09) * (5.508 \times 10^{-3}) = 6.004 \times 10^{-3} \text{ ft/ft for } \frac{W}{H} = 0.62$$

Max tensile strain = $6.108 \times 10^{-3} \text{ ft/ft}$

Max Compressive strain = $8.997 \times 10^{-3} \text{ ft/ft}$

5. Three geophones are located along a tunnel at 10 m (G1), 310 m (G2), and 640 m (G3) from the tunnel entrance. Longitudinal wave velocity in the adjacent rock mass is 3,000 m/s. An event arrival is detected at G1 at 10:31:20.06 (hours, minutes, seconds). Event arrivals are also detected at G2 at 10:31:20.16 and at G3 at 10:31:20.27. Determine if the event is locatable and if so, best estimates of location and time of the event. **(2 marks)**

Solution: Time taken by wave to travel from G1 to G2 = 0.10 s [10:31:20.06 + 0.10 = 10:31:20.16]

Time taken by wave to travel from G2 to G3 = 0.11 s [10:31:20.16+0.11 = 10.:31:20.27]

Therefore, event occurred at G1 at 10:31.20.06.

6. A vertical shaft of elliptical section is being considered in a region of high horizontal stress. Measurements along the shaft route indicate the preshaft principal stresses in psi (compression positive) are given by $S_v = 1.1H$, $S_h = 100 + 1.5H$, $S_H = 500 + 2.2H$ (H is depth in feet). Determine the optimum orientation and aspect ratio of the section.

$$K_a = -1 + M \left(\frac{2}{k} + 1 \right)$$

$$K_b = (1 + 2k) - M$$

(4 marks)

Solution: The peak stress concentrations at the wall of an elliptical section occur at the ends of the geometric axes.

The peak stresses are given by the formulas

$$\sigma_a = K_a S_1 = -S_1 + \left(1 + \frac{2}{k} \right) S_3 = S_1 \left[\left(1 + \frac{2}{k} \right) M - 1 \right]$$

$$\sigma_b = K_b S_1 = -S_3 + (1 + 2k) S_1 = S_1 [(1 + 2k) - M]$$

where the meaning of terms is shown in the sketch, $k = \frac{b}{a}$ and $M = \frac{S_3}{S_1}$.

An optimum orientation would be one that reduces stress concentration to a minimum, so the stress concentration at b is no greater or less than at a and tension is absent.

Thus, $K_a = K_b \geq 0$

$$\therefore \left(1 + \frac{2}{k} \right) M - 1 = (1 + 2k) - M \therefore k = M = \frac{100 + 1.5H}{500 + 2.2H}$$

so the aspect ratio varies with depth, but is equal to the preexcavation stress ratio at any particular depth.

At the surface $k = 0.200$ and at great depth $k = \frac{1.5}{2.2} = 0.682$. The best orientation is with the long axis parallel to S_H .

In this orientation, with $k = M$, $K_a = K_b = 1 + M$ and indeed the wall of the ellipse is in a uniform compression. At the surface, the stress concentration is 1.2; at great depth the stress concentration is 1.682.

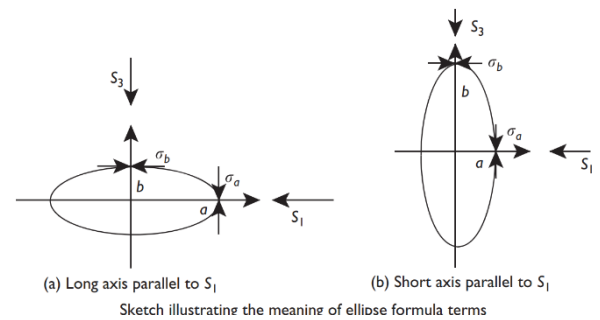
With the short axis parallel to S_H , $K'_a = 2M^2 + M$, and $K'_b = 1 - M + \frac{2}{M}$, which is greater than $1 + M$ and therefore less favorable at any depth.

If a circular section were considered, the stress concentrations would be $-1 + 3M$ and $3 - M$ or -0.4 and 2.8 at the surface; 1.046 and 2.318 at great depth.

7. A rectangular shaft $3.7 \text{ m} \times 7.4 \text{ m}$ is planned for deepening from a depth of 975 m to $1,463 \text{ m}$ in ground where the preshaft stress state relative to compass coordinates ($x = \text{east}$, $y = \text{north}$, $z = \text{up}$) is given (in kPa) by $S_{xx} = 13793 + 24.9H$, $S_{yy} = 345 + 20.4H$, $S_{zz} = 26.0H$, $T_{xy} = -2414 - 6.8H$, $T_{yz} = 0.0$, $T_{zx} = 0.0$, where compression is positive and H is depth in m below surface.

(i) Find the best orientation of the shaft at a planned $1,175 \text{ m}$ deep shaft station and show by sketch. (ii) Also show on the sketch where the peak stress concentrations are likely to occur. (iii) Finally, estimate the peak stress concentrations.

Table: Stress concentration factors about a rectangular section.



$\frac{k}{k'}$	M										
	0		1/5		1/3		1/2		3/4		1
	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$K_{\max}$ $K_{\min}$
1/8	3.70	-0.84	5.30	-0.18	6.62	0.20	8.28	0.64	11.1	0.42	14.04
8	11.8	-1.01	12.08	-0.77	12.4	-0.60	12.8	-0.40	13.4	-0.10	0.18
1/4	2.96	-0.81	3.72	-0.29	4.36	0.03	5.28	0.42	6.80	0.56	8.43
4	6.96	-0.96	7.21	-0.71	7.37	-0.53	7.58	-0.32	7.99	0.00	0.32
1/2	3.28	-0.82	3.70	-0.41	4.05	-0.14	4.57	0.19	5.48	0.69	6.46
2	4.72	-0.92	4.97	-0.64	5.15	-0.44	5.44	-0.20	5.92	0.16	0.52
1	3.58	-0.85	3.83	-0.52	4.06	-0.29	4.39	-0.02	4.97	0.40	5.65
											0.80

$$\sigma_{xx} = 13793 + 24.9H = 13793 + 24.9 * (1175) = 43.05 \text{ MPa}$$

$$\sigma_{yy} = 345 + 20.4H = 345 + 20.4 * (1175) = 24.32 \text{ MPa}$$

$$\sigma_{zz} = 26.0H = 26.0 * (1175) = 30.55 \text{ MPa}$$

$$\tau_{xy} = -2414 - 6.8H = -2414 - 6.8 * (1175) = -10.40 \text{ MPa}$$

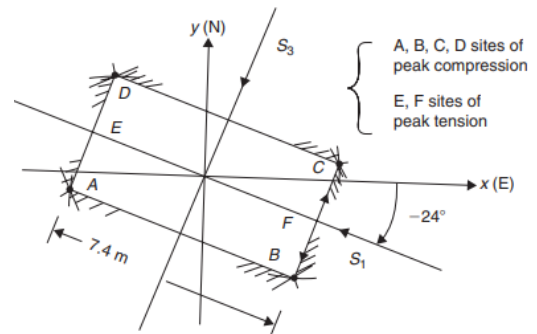
$$\tau_{yz} = 0.0, \tau_{zx} = 0.0$$

z is a principal direction, by inspection

Orientation of the major principal stress in the xy-plane is given by

$$\tan(2\theta) = \frac{\tau_{xy}}{\frac{(\sigma_{xx} - \sigma_{yy})}{2}} = \frac{-10.40}{\frac{(43.05 - 24.32)}{2}} = -1.1105$$

$$\theta = -24^\circ, 66^\circ$$



$$S_1 = \left(\frac{\sigma_{xx} + \sigma_{yy}}{2} \right) \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2} \right)^2 + (\tau_{xy})^2} = \left(\frac{43.05 + 24.32}{2} \right) \pm \sqrt{\left(\frac{43.05 - 24.32}{2} \right)^2 + (-10.40)^2}$$

$$= 33.69 \pm 14$$

$$S_1 = 47.69 \text{ MPa and } S_3 = 19.69 \text{ MPa}$$

$$M = \frac{S_3}{S_1} = \frac{19.69}{47.69} = 0.412 = M$$

$$\frac{W_o}{H_o} = k = \frac{1}{2} = 0.5$$

$M = \frac{1}{3}$	$M = \frac{1}{2}$	$k = \frac{1}{2}$
$K_{\max}$	4.05	4.57
$K_{\min}$	-0.14	-0.20

at $M = 0.412$ $K_{\max} = 4.31$ (interpolate) $K_{\min} = -0.17$

$$\sigma_c = K_c \sigma_1 = 4.31 * 43.05 = 185.5 \text{ MPa} \leftarrow$$

$$\sigma_t = K_t \sigma_1 = -0.17 * 43.05 = -7.31 \text{ MPa} \leftarrow$$

OR

Consider a vertical shaft excavated in a stress field where $S_v = 1,350$, S_h (north) = 300 psi and S_H (east) = 1,200 psi and a joint set that strikes due north and dips 60° with cohesion $c_j = 135$ psi and friction angle $\phi_j = 35^\circ$. For convenience, fix the x-axis due east parallel with S_H . Determine joint safety factors before excavation, after excavation and the improvement in safety factor that a bolting pattern could achieve by installing horizontally in square grid pattern of 3 ft centers and tensioned at 18500 pounds.

$$\Delta FS_j = \frac{F_b \sin(\delta - \beta) \tan(\phi_j) + F_b \cos(\delta - \beta)}{\tau_j(stress)A_j}$$

(5 marks)

Solution: Joint normal and shear stresses for this purpose may be obtained by rotation of the reference axes.

Thus, before excavation

$$\sigma_j = \left(\frac{1200 + 1350}{2} \right) - \left(\frac{1200 - 1350}{2} \right) \cos(120^\circ) = 1237.5 \text{ psi}$$

$$\tau_j = - \left(\frac{1200 - 1350}{2} \right) \sin(120^\circ) = 65 \text{ psi}$$

where the presence of the joint is assumed not to affect the given stress field.

The joint safety factor is now

$$FS_j(before) = \frac{1237.5 \tan(35^\circ) + 135}{65} = 15.40$$

so joint slip or separation is not impending. After excavation

$$\sigma_j = \left(\frac{0 + 1350}{2} \right) - \left(\frac{0 - 1350}{2} \right) \cos(120^\circ) = 337.5 \text{ psi}$$

$$\tau_j = - \left(\frac{0 - 1350}{2} \right) \sin(120^\circ) = 585 \text{ psi}$$

which shows some normal stress reduction but a significant increase in shear stress; both tend to reduce safety. The joint safety factor is now

$$FS_j(after) = \frac{337.5 \tan(35^\circ) + 135}{585} = 0.6347$$

which is greatly reduced and probably too low for shaft wall safety.

If bolts are installed horizontally on a square pattern of three-foot centers and tensioned to 18,500 pounds force, the safety factor improvement is

$$\Delta FS_j = \frac{18500[\sin(60^\circ - 0) \tan(35^\circ) + \cos(60^\circ - 0)]}{(585)(3)(3) \frac{\cos(0)}{\sin(60^\circ)}} = 0.023$$

or less than 2% over the post-excavation joint safety factor

*****End of the Question Paper*****